1. Fubini's theorem states that for a continuous function f(x, y) on a rectangle [a, b] x [c, d], the double integral can be evaluated as:

(A) An ordinary derivative
(B) An iterated integral in either order of integration
(C) A single Riemann sum
(D) Zero

2. The area of a closed plane region R is given by the double integral:

(A) iint_R x dA
(B) iint_R 1 dA
(C) iint_R y dA
(D) iint_R (x + y) dA

3. When converting a double integral from Cartesian to polar coordinates, the area element dx dy is replaced by:

(A) dr dtheta
(B) r dr dtheta
(C) r^2 dr dtheta
(D) (1/r) dr dtheta

4. The volume of a 3D solid region V bounded below by z = 0 and above by z = f(x, y) >= 0 over region R is:

(A) iint_R f(x, y) dA
(B) iint_R [f(x, y)]^2 dA
(C) int_R f(x, y) dx
(D) 2*pi * iint_R f(x, y) dA

5. In changing the order of integration of int_0^1 int_0^x f(x, y) dy dx, the new limits are:

(A) int_0^1 int_0^y f dx dy
(B) int_0^1 int_y^1 f dx dy
(C) int_0^x int_0^1 f dx dy
(D) int_0^1 int_0^1 f dx dy

6. The mass M of a planar lamina R with variable surface density rho(x, y) is:

(A) iint_R 1 dA
(B) iint_R rho(x, y) dA
(C) iint_R x*rho dA
(D) iint_R y*rho dA

7. The x-coordinate of the center of mass (centroid) of a lamina R with uniform density is:

(A) iint_R y dA / Area
(B) iint_R x dA / Area
(C) iint_R x^2 dA / Area
(D) Area / iint_R x dA

8. The moment of inertia I_x of a lamina R with density rho about the x-axis is:

(A) iint_R x^2 * rho dA
(B) iint_R y^2 * rho dA
(C) iint_R (x^2 + y^2) * rho dA
(D) iint_R x*y * rho dA

9. The polar moment of inertia I_0 about the origin satisfies the perpendicular axis theorem:

(A) I_0 = I_x - I_y
(B) I_0 = I_x + I_y
(C) I_0 = I_x * I_y
(D) I_0 = sqrt(I_x^2 + I_y^2)

10. In cylindrical coordinates (r, theta, z), the differential volume element dV is:

(A) dr dtheta dz
(B) r dr dtheta dz
(C) r^2 dr dtheta dz
(D) r * sin theta dr dtheta dz

11. In spherical polar coordinates (rho, theta, phi), the differential volume element dV is:

(A) rho drho dtheta dphi
(B) rho^2 * sin theta drho dtheta dphi
(C) rho^2 * cos theta drho dtheta dphi
(D) rho^3 drho dtheta dphi

12. The volume of a sphere of radius R evaluated by triple integration is:

(A) 2 * pi * R^3
(B) (4/3) * pi * R^3
(C) pi * R^3
(D) (2/3) * pi * R^3

13. A line integral int_C F . dr represents the physical work done by a force field F along curve C if F and r are:

(A) Charge and distance
(B) Force vector and displacement differential
(C) Velocity and acceleration
(D) Pressure and volume

14. A vector field F is called conservative if there exists a scalar potential function phi such that:

(A) F = curl(phi)
(B) F = grad(phi)
(C) div(F) = 0
(D) F . phi = 0

15. If a force field F is conservative, the line integral int_C F . dr between two fixed points A and B is:

(A) Dependent on the exact path taken
(B) Independent of the path connecting A and B
(C) Always zero
(D) Infinite

16. The line integral of a conservative vector field around ANY closed path C (oint_C F . dr) is:

(A) 1
(B) 0
(C) 2*pi
(D) Area enclosed

17. A necessary and sufficient condition for a continuous vector field F in R^3 to be conservative in a simply connected domain is:

(A) div(F) = 0
(B) curl(F) = 0 (F is irrotational)
(C) div(F) = 1
(D) grad(F) = 0

18. Green's Theorem in a plane relates a line integral around a closed boundary curve C to a:

(A) Single integral along the normal
(B) Double integral over the enclosed plane region R
(C) Triple integral over a sphere
(D) Surface integral in R^3

19. The mathematical statement of Green's Theorem is oint_C (P dx + Q dy) equals:

(A) iint_R (dP/dy + dQ/dx) dA
(B) iint_R (dQ/dx - dP/dy) dA
(C) iint_R (dP/dx - dQ/dy) dA
(D) iint_R (dQ/dy - dP/dx) dA

20. Using Green's Theorem, the area of a closed plane region R bounded by C is given by the line integral:

(A) oint_C (x dx + y dy)
(B) (1/2) * oint_C (x dy - y dx)
(C) oint_C (x dy + y dx)
(D) (1/2) * oint_C (x dx - y dy)

21. Stokes' Theorem relates the line integral of vector field F around a closed curve C to the:

(A) Volume integral of div(F)
(B) Surface integral of curl(F) over any surface S bounded by C
(C) Double integral of grad(F)
(D) Flux of F itself

22. The mathematical formulation of Stokes' Theorem is oint_C F . dr equals:

(A) iint_S (curl F) . n dS
(B) iint_S (div F) dS
(C) iiint_V (curl F) dV
(D) oint_C (div F) ds

23. If curl(F) = 0 everywhere on surface S, then by Stokes' Theorem, the circulation oint_C F . dr is:

(A) 1
(B) 0
(C) Area of S
(D) pi

24. Gauss' Divergence Theorem transforms a surface integral of flux over a closed surface S into a:

(A) Line integral around boundary
(B) Triple integral of the divergence of F over the enclosed volume V
(C) Double integral of curl F
(D) Gradient integral

25. The mathematical formulation of Gauss' Divergence Theorem is iint_S F . n dS equals:

(A) iiint_V (curl F) dV
(B) iiint_V (div F) dV
(C) oint_C F . dr
(D) iint_S (grad F) dS

26. A vector field F is called solenoidal if its divergence satisfies:

(A) curl(F) = 0
(B) div(F) = 0
(C) grad(F) = 0
(D) div(F) > 0

27. If a vector field F is solenoidal throughout a volume V bounded by closed surface S, the net outward flux iint_S F . n dS is:

(A) 1
(B) 0
(C) Volume V
(D) Surface area S

28. In biological transport phenomena, Fick's first law defines solute flux J = -D * grad(C). The net mass flux out of a cell boundary S is:

(A) iint_S J . n dS
(B) iiint_V C dV
(C) oint_C C ds
(D) div(J)

29. In an incompressible fluid (such as culture broth in a bioreactor), the velocity vector field v satisfies:

(A) curl(v) = 0
(B) div(v) = 0 (solenoidal)
(C) v . grad(v) = 1
(D) div(v) > 0

30. The value of the double integral int_0^1 int_0^1 (x + y) dx dy is:

(A) 1/2
(B) 1
(C) 2
(D) 1/4

31. Evaluate int_0^1 int_0^1 x * y dx dy:

(A) 1/2
(B) 1/4
(C) 1
(D) 1/8

32. The value of int_0^2 int_0^3 1 dx dy is simply the area of the rectangle, equal to:

(A) 5
(B) 6
(C) 12
(D) 1

33. Dirichlet's integral formula evaluates multiple integrals over regions bounded by:

(A) x >= 0, y >= 0, z >= 0 and x + y + z <= 1
(B) Circles
(C) Hyperbolas
(D) Paraboloids

34. According to Dirichlet's formula, iiint_V x^(l-1) y^(m-1) z^(n-1) dx dy dz over x+y+z<=1 is:

(A) [Gamma(l)*Gamma(m)*Gamma(n)] / Gamma(l+m+n+1)
(B) [Gamma(l)+Gamma(m)+Gamma(n)] / Gamma(l+m+n)
(C) Gamma(l*m*n)
(D) 1 / (l*m*n)

35. The surface area of a smooth surface z = f(x, y) over region R in the xy-plane is:

(A) iint_R sqrt(1 + (dz/dx)^2 + (dz/dy)^2) dA
(B) iint_R [1 + dz/dx + dz/dy] dA
(C) iint_R z dA
(D) iint_R sqrt(x^2 + y^2 + z^2) dA

36. Evaluate div(r) where r = x*i + y*j + z*k is the position vector in R^3:

(A) 0
(B) 1
(C) 3
(D) r

37. Evaluate curl(r) where r = x*i + y*j + z*k is the position vector:

(A) 0 (Zero vector)
(B) 3
(C) 1
(D) 2*r

38. The divergence of the curl of any twice continuously differentiable vector field div(curl F) is:

(A) 1
(B) 0 identically
(C) -1
(D) grad(div F)

39. The curl of the gradient of any twice continuously differentiable scalar field curl(grad phi) is:

(A) 0 identically
(B) 1
(C) Laplacian of phi
(D) grad(div phi)

40. The Laplacian operator del^2 phi is defined as:

(A) div(curl phi)
(B) div(grad phi)
(C) curl(grad phi)
(D) grad(div phi)

41. Evaluate int_0^1 int_0^(x^2) x * y dy dx:

(A) 1/12
(B) 1/6
(C) 1/24
(D) 1/4

42. Evaluate int_0^(pi/2) int_0^a r * dr dtheta (area of a quarter circle of radius a):

(A) pi * a^2
(B) pi * a^2 / 2
(C) pi * a^2 / 4
(D) 2 * pi * a^2

43. Evaluate int_0^infinity int_0^infinity e^(-(x^2 + y^2)) dx dy by converting to polar coordinates:

(A) pi
(B) pi / 2
(C) pi / 4
(D) sqrt(pi)

44. Change the order of integration for I = int_0^a int_x^a (x^2 + y^2) dy dx. The new integral is:

(A) int_0^a int_0^y (x^2 + y^2) dx dy
(B) int_0^a int_y^a (x^2 + y^2) dx dy
(C) int_x^a int_0^a (x^2 + y^2) dx dy
(D) int_0^a int_0^a (x^2 + y^2) dx dy

45. Evaluate int_0^1 int_y^1 e^(x^2) dx dy by reversing the order of integration:

(A) (e - 1) / 2
(B) e - 1
(C) (e + 1) / 2
(D) e / 2

46. Evaluate int_0^pi int_0^(a*sin theta) r dr dtheta:

(A) pi * a^2 / 2
(B) pi * a^2 / 4
(C) pi * a^2
(D) a^2 / 2

47. Find the area bounded by the cardioid r = a*(1 + cos theta) using double integration:

(A) (3/2) * pi * a^2
(B) pi * a^2
(C) 2 * pi * a^2
(D) 3 * pi * a^2

48. Evaluate the triple integral int_0^1 int_0^1 int_0^1 (x + y + z) dx dy dz:

(A) 1/2
(B) 1
(C) 3/2
(D) 2

49. Evaluate the volume of the tetrahedron bounded by coordinate planes and x + y + z = 1:

(A) 1/2
(B) 1/6
(C) 1/3
(D) 1/24

50. The mass of the solid ellipsoid x^2/a^2 + y^2/b^2 + z^2/c^2 <= 1 with uniform density rho0 is:

(A) 4 * pi * a * b * c * rho0
(B) (4/3) * pi * a * b * c * rho0
(C) (2/3) * pi * a * b * c * rho0
(D) pi * a * b * c * rho0

51. Evaluate the line integral int_C (x^2 * dx + x*y * dy) from (0, 0) to (1, 1) along the straight line y = x:

(A) 1/3
(B) 2/3
(C) 5/6
(D) 1

52. Evaluate the line integral int_C (x^2 * dx + x*y * dy) from (0, 0) to (1, 1) along the parabola y = x^2:

(A) 1/3 + 2/5 = 11/15
(B) 7/12
(C) 5/6
(D) 2/3

53. The force field F = (2xy + z^3)*i + x^2*j + 3xz^2*k has potential function phi equal to:

(A) x^2 * y + x * z^3 + C
(B) 2xy + z^3 + C
(C) x^2 * y * z + C
(D) x*y*z + C

54. Evaluate the work done by conservative force F = grad(x^2 * y) in moving a particle from (1, 1) to (2, 3):

(A) 12 - 1 = 11
(B) 10
(C) 8
(D) 14

55. Verify Green's Theorem for oint_C (x*y + y^2)*dx + x^2*dy where C is closed curve bounded by y = x and y = x^2. The value is:

(A) 1/20
(B) -1/20
(C) 1/12
(D) -1/12

56. Using Green's theorem, compute the area of the ellipse x = a*cos t, y = b*sin t:

(A) pi * a * b
(B) 2 * pi * a * b
(C) 4 * pi * a * b
(D) pi * (a^2 + b^2)

57. Evaluate oint_C (y * dx - x * dy) around the unit circle x^2 + y^2 = 1 traversed counter-clockwise:

(A) 0
(B) -2*pi
(C) 2*pi
(D) pi

58. If F = y*i + z*j + x*k, find curl(F):

(A) -i - j - k
(B) i + j + k
(C) 0
(D) 2*(i + j + k)

59. Apply Stokes' theorem to evaluate oint_C F . dr where F = y*i + z*j + x*k and C is circle x^2 + y^2 = 1, z = 0:

(A) -pi
(B) pi
(C) 0
(D) 2*pi

60. Evaluate iint_S r . n dS over closed surface S bounding volume V where r = x*i + y*j + z*k:

(A) V
(B) 2 * V
(C) 3 * V (by Gauss Divergence Theorem)
(D) 0

61. Evaluate the outward flux of F = 2x*i + 3y*j + z*k through the surface of unit sphere x^2 + y^2 + z^2 = 1:

(A) 6 * (4/3)*pi = 8*pi
(B) 4*pi
(C) 2*pi
(D) 12*pi

62. Evaluate the line integral oint_C (3x + 4y)*dx + (2x - 3y)*dy around circle x^2 + y^2 = 4 using Green's Theorem:

(A) -2 * (4*pi) = -8*pi
(B) 8*pi
(C) 4*pi
(D) 0

63. Evaluate the surface integral iint_S (x*i + y*j + z*k) . n dS over the cube [0, 1] x [0, 1] x [0, 1]:

(A) 1
(B) 2
(C) 3
(D) 6

64. A scalar potential exists for vector field F if and only if:

(A) div(F) = 0
(B) curl(F) = 0
(C) div(F) = curl(F)
(D) F . dr = 0

65. In an immobilized enzyme spherical bead of radius R, nutrient concentration satisfies del^2 C = (1/r^2)*d/dr(r^2 dC/dr) = k/D. In steady state, total consumption rate within the bead is:

(A) iiint_V k dV = (4/3)*pi*R^3 * k
(B) 4*pi*R^2 * k
(C) k * R
(D) 2*pi*R^3 * k

66. The surface area of the portion of plane x + y + z = a in the first octant is:

(A) sqrt(3) * a^2 / 2
(B) a^2 / 2
(C) sqrt(2) * a^2
(D) 3 * a^2

67. The moment of inertia of a uniform solid sphere of mass M and radius R about its diameter is:

(A) (2/5) * M * R^2
(B) (1/2) * M * R^2
(C) (2/3) * M * R^2
(D) (3/5) * M * R^2

68. The centroid (x_bar, y_bar) of the region bounded by y = x^2 and y = 4 has y_bar equal to:

(A) 2.0
(B) 2.4 = 12/5
(C) 2.5
(D) 3.0

69. Evaluate int_0^1 int_0^sqrt(1 - x^2) dy dx (area of quarter unit disk):

(A) pi / 2
(B) pi / 4
(C) pi
(D) 1

70. Evaluate int_0^a int_0^b int_0^c dx dy dz:

(A) a + b + c
(B) a * b * c
(C) (1/6) * abc
(D) (1/3) * abc

71. The curl of the velocity field in a rigid body rotation v = omega x r is equal to:

(A) omega
(B) 2 * omega
(C) 0
(D) -omega

72. If F is the gradient of a harmonic function (del^2 phi = 0), then F is:

(A) Both solenoidal and irrotational
(B) Solenoidal only
(C) Irrotational only
(D) Rotational

73. The circulation of fluid velocity v around a closed contour C is defined as:

(A) oint_C v . dr
(B) iint_S v . n dS
(C) div(v)
(D) grad(v)

74. Gauss's law in electrostatics relates total electric flux through a closed surface to:

(A) Enclosed electric charge Q_enc / epsilon0
(B) Magnetic field
(C) Electric potential
(D) Current

75. The value of iint_S curl(F) . n dS over ANY closed surface S is identically:

(A) 1
(B) 0 (since a closed surface has no boundary curve)
(C) Volume enclosed
(D) 4*pi

76. Evaluate by changing order of integration: int_0^infinity int_x^infinity (e^(-y) / y) dy dx:

(A) 1
(B) 0
(C) e
(D) 1/2

77. Evaluate int_0^1 int_0^sqrt(1 - x^2) (x^2 + y^2) dy dx by converting to polar coordinates:

(A) pi / 8
(B) pi / 4
(C) pi / 16
(D) pi / 2

78. Evaluate the volume of the region common to the two intersecting cylinders x^2 + y^2 = a^2 and x^2 + z^2 = a^2 (Steinmetz solid):

(A) 16*a^3 / 3
(B) 8*a^3 / 3
(C) 4*pi*a^3 / 3
(D) 8*pi*a^3

79. Evaluate iiint_V (x + y + z) dx dy dz over the region bounded by x=0, y=0, z=0 and x + y + z = 1:

(A) 1 / 8
(B) 1 / 24
(C) 1 / 12
(D) 1 / 4

80. Evaluate the integral iiint_V z dV over the upper hemisphere x^2 + y^2 + z^2 <= R^2 (z >= 0):

(A) pi * R^4 / 4
(B) pi * R^4 / 2
(C) (4/3)*pi*R^4
(D) pi * R^3 / 3

81. The z-coordinate of the center of mass of a uniform solid hemisphere of radius R is:

(A) 3R / 8
(B) R / 2
(C) 5R / 8
(D) 3R / 4

82. Evaluate the line integral oint_C [(y - sin x)*dx + cos x * dy] where C is the triangle bounded by y = 0, x = pi/2, y = (2/pi)*x using Green's Theorem:

(A) pi / 4 - 1
(B) -(pi / 4)
(C) - (1/2)
(D) 1 - pi/4

83. Verify Stokes' Theorem for F = (2x - y)*i - y*z^2*j - y^2*z*k over the upper half-sphere x^2 + y^2 + z^2 = 1 (z >= 0). The value of circulation is:

(A) pi
(B) 2 * pi
(C) 0
(D) -pi

84. Evaluate iint_S F . n dS using Gauss Divergence Theorem for F = 4x*i - 2y^2*j + z^2*k over the cylinder x^2 + y^2 = 4, z = 0, z = 3:

(A) 84 * pi
(B) 48 * pi
(C) 96 * pi
(D) 72 * pi

85. Evaluate the volume enclosed by the paraboloid z = x^2 + y^2 and the plane z = 4:

(A) 8 * pi
(B) 16 * pi
(C) 4 * pi
(D) 32 * pi

86. Evaluate the triple integral iiint_V sqrt(x^2 + y^2 + z^2) dx dy dz over the sphere x^2 + y^2 + z^2 <= a^2:

(A) pi * a^4
(B) 2 * pi * a^4
(C) (4/3) * pi * a^4
(D) (1/2) * pi * a^4

87. Using Dirichlet's theorem, evaluate iiint_V x^(1/2) y^(1/2) z^(1/2) dx dy dz over the tetrahedron x+y+z<=1 (x, y, z >= 0):

(A) pi * sqrt(pi) / 210
(B) pi^2 / 105
(C) sqrt(pi) / 70
(D) pi / 140

88. Evaluate the line integral int_C (y^2 * dx + 2xy * dy + z * dz) from (0, 0, 0) to (1, 1, 1):

(A) 3 / 2
(B) 1
(C) 2
(D) 1/2

89. Find the flux of F = x^3*i + y^3*j + z^3*k across the surface of the sphere x^2 + y^2 + z^2 = a^2:

(A) (12/5) * pi * a^5
(B) (4/5) * pi * a^5
(C) 4 * pi * a^5
(D) (3/5) * pi * a^5

90. In bioreactor gas sparging, oxygen bubble dissolution flux J_O2 = kL * (C_sat - C). For a spherical bubble of radius R, total oxygen transfer rate per bubble is:

(A) 4 * pi * R^2 * kL * (C_sat - C)
(B) 4/3 * pi * R^3 * kL
(C) 2 * pi * R * kL
(D) pi * R^2 * kL

91. Evaluate the line integral oint_C (e^x * dx + 2y * dy - dz) around any closed curve C in space:

(A) 0 (since F is conservative)
(B) 1
(C) 2*pi
(D) pi

92. Evaluate the surface area of the portion of cylinder x^2 + z^2 = a^2 inside the cylinder x^2 + y^2 = a^2:

(A) 8 * a^2
(B) 4 * a^2
(C) 16 * a^2
(D) 2 * pi * a^2

93. The centroid of the area in the first quadrant bounded by the astroid x^(2/3) + y^(2/3) = a^(2/3) has coordinates (x_bar, y_bar):

(A) (256*a / (315*pi), 256*a / (315*pi))
(B) (3a/8, 3a/8)
(C) (a/2, a/2)
(D) (4a/3pi, 4a/3pi)

94. Evaluate the double integral iint_R (x^2 + y^2) dA over the region R bounded by y = x^2 and y = x:

(A) 3 / 35
(B) 1 / 35
(C) 2 / 35
(D) 4 / 35

95. Evaluate the volume of the solid bounded by the coordinate planes and the surface sqrt(x) + sqrt(y) + sqrt(z) = 1:

(A) 1 / 90
(B) 1 / 60
(C) 1 / 120
(D) 1 / 30

96. Compute the circulation of F = (x - y)*i + (x + y)*j around the square with vertices (+-1, +-1) using Green's Theorem:

(A) 8
(B) 4
(C) 2
(D) 0

97. The scalar potential phi for F = (y^2*z^3)*i + (2xyz^3)*j + (3xy^2*z^2)*k with phi(0, 0, 0) = 0 is:

(A) x * y^2 * z^3
(B) x^2 * y * z^3
(C) 3 * x * y * z
(D) x^2 + y^2 + z^2

98. Evaluate iint_S (x*dy dz + y*dz dx + z*dx dy) over the outer surface of the ellipsoid x^2/a^2 + y^2/b^2 + z^2/c^2 = 1:

(A) 4 * pi * a * b * c
(B) 3 * pi * a * b * c
(C) (4/3) * pi * a * b * c
(D) 2 * pi * a * b * c

99. Evaluate the line integral oint_C (z dx + x dy + y dz) along the intersection of x^2 + y^2 + z^2 = a^2 and x + y + z = 0 using Stokes' Theorem:

(A) pi * a^2 * sqrt(3)
(B) pi * a^2
(C) -pi * a^2 * sqrt(3)
(D) 0

100. In cell suspension fluid mechanics, if vorticity omega = curl(v) = 0 everywhere in an irrotational vortex flow, the velocity circulation around any closed loop not enclosing the core is:

(A) 0
(B) 1
(C) 2*pi
(D) Infinity

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Explain Fubini's Theorem for double integrals over rectangular regions. Evaluate the double integral: I = iint_R (x^2 + y^2) dA over the rectangle R = [0, 2] x [0, 3] in both orders of integration.

Q2. Describe the methodology for changing the order of integration in a double integral. Change the order of integration and evaluate: I = int_0^a int_{x^2/a}^{2a - x} x * y dy dx.

Q3. Change the order of integration and evaluate the non-elementary integral: I = int_0^1 int_y^1 e^(x^2) dx dy.

Q4. Derive the polar coordinate transformation for double integrals. Use polar coordinates to evaluate the famous improper integral: I = int_0^infinity int_0^infinity e^(-(x^2 + y^2)) dx dy, and hence deduce the value of the Gaussian integral int_0^infinity e^(-x^2) dx.

Q5. Evaluate the double integral: I = iint_R sqrt(a^2 - x^2 - y^2) dx dy, where R is the region in the first quadrant enclosed by the circle x^2 + y^2 = a^2, by transforming into polar coordinates.

Q6. Using double integration, find the area enclosed between the parabola y = x^2 and the straight line y = 2x.

Q7. Using double integration in polar coordinates, calculate the area of the region bounded by the cardioid r = a * (1 + cos theta).

Q8. Derive formulas for the center of mass (centroid) and moments of inertia of a plane lamina R using double integration. Find the centroid of the region bounded by the parabola y^2 = 4ax and its latus rectum x = a.

Q9. Find the moment of inertia of a uniform circular disc of mass M and radius R about: (i) An axis perpendicular to the disc through its center (polar moment of inertia), and (ii) Any diameter.

Q10. Evaluate the volume of the tetrahedron bounded by the coordinate planes x = 0, y = 0, z = 0 and the plane x + y + z = 1 using triple integration.

Q11. Explain cylindrical coordinates and evaluate the volume of the solid enclosed by the paraboloid z = x^2 + y^2 and the horizontal plane z = 4.

Q12. Explain spherical polar coordinates (rho, theta, phi) and evaluate the triple integral: I = iiint_V sqrt(x^2 + y^2 + z^2) dx dy dz over the sphere x^2 + y^2 + z^2 <= a^2.

Q13. State Dirichlet's Theorem for triple integrals. Use it to evaluate: I = iiint_V x^(l-1) * y^(m-1) * z^(n-1) dx dy dz over the positive octant bounded by x + y + z <= 1 (x, y, z >= 0).

Q14. Find the volume common to the two intersecting cylinders of equal radius a: x^2 + y^2 = a^2 and x^2 + z^2 = a^2 (Steinmetz solid).

Q15. Define a vector line integral and evaluate the work done by the force field F = (3x^2 + 6y)*i - 14yz*j + 20xz^2*k in moving a particle from (0, 0, 0) to (1, 1, 1) along the parametric curve x = t, y = t^2, z = t^3.

Q16. Define a conservative vector field. Prove that F = (2xy + z^3)*i + x^2*j + 3xz^2*k is conservative, find its scalar potential function phi(x, y, z), and evaluate the work done in moving from (1, -2, 1) to (3, 1, 4).

Q17. State Green's Theorem in a plane. Provide a complete analytical proof for a simple region R bounded by a piecewise smooth curve C.

Q18. Verify Green's Theorem for the line integral oint_C [(x*y + y^2)*dx + x^2*dy], where C is the closed curve bounded by y = x and y = x^2.

Q19. Derive the line integral area formula from Green's Theorem: Area = (1/2) * oint_C (x dy - y dx). Apply this formula to find the area enclosed by the astroid: x = a*cos^3(t), y = a*sin^3(t).

Q20. State Stokes' Theorem for a vector field F. Give its physical interpretation and state how it reduces to Green's Theorem in the xy-plane.

Q21. Verify Stokes' Theorem for F = (2x - y)*i - y*z^2*j - y^2*z*k over the upper hemisphere surface x^2 + y^2 + z^2 = 1 (z >= 0), bounded by the circle x^2 + y^2 = 1 in the plane z = 0.

Q22. State Gauss' Divergence Theorem. Provide a rigorous analytical proof for a simple 3D volume V bounded by an orientable closed surface S.

Q23. Verify Gauss' Divergence Theorem for the vector field F = 4x*i - 2y^2*j + z^2*k over the cylinder x^2 + y^2 = 4, bounded by z = 0 and z = 3.

Q24. Use Gauss' Divergence Theorem to evaluate the outward flux: iint_S (x^3*i + y^3*j + z^3*k) . n dS over the spherical surface S: x^2 + y^2 + z^2 = a^2.

Q25. Evaluate the surface integral iint_S F . n dS, where F = z*i + x*j - 3y^2*z*k and S is the surface of the cylinder x^2 + y^2 = 16 included in the first octant between z = 0 and z = 5.

Q26. Evaluate the line integral oint_C (z dx + x dy + y dz) along the intersection of the sphere x^2 + y^2 + z^2 = a^2 and the plane x + y + z = 0 using Stokes' Theorem.

Q27. Prove that the field of flow of an incompressible fluid has zero divergence (div v = 0) from the law of conservation of mass. Formulate this via Gauss' Divergence Theorem.

Q28. In biotechnology bioreactor design, use surface flux integration to calculate the total oxygen mass transfer rate across spherical gas bubbles of uniform radius R = 1 mm. Given gas holdup epsilon_g = 0.1, liquid volume V_L = 10 m^3, mass transfer coefficient k_L = 4 * 10^-4 m/s, and

Q29. In an immobilized enzyme spherical bead of radius R, substrate diffuses radially and undergoes zero-order reaction: (D / r^2) * d/dr (r^2 * dC/dr) = k0. Integrate this second-order ODE in spherical coordinates subject to boundary conditions dC/dr(0) = 0 and C(R) = Cs, and determine the

Q30. Use the Divergence Theorem to show that for a closed cellular surface S bounding cell volume V, the total diffusive flux of a metabolite governed by Fick's law J = -D * grad(C) equals the total volumetric reaction rate iiint_V R_rxn dV at steady state.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[B]	**Q21:**	[B]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[B]	**Q22:**	[A]	**Q42:**	[C]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[B]	**Q23:**	[B]	**Q43:**	[C]	**Q63:**	[C]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[B]	**Q44:**	[A]	**Q64:**	[B]	**Q84:**	[A]
Q5:	[B]	**Q25:**	[B]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[B]	**Q26:**	[B]	**Q46:**	[B]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[B]	**Q27:**	[B]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[B]	**Q28:**	[A]	**Q48:**	[C]	**Q68:**	[B]	**Q88:**	[A]
Q9:	[B]	**Q29:**	[B]	**Q49:**	[B]	**Q69:**	[B]	**Q89:**	[A]
Q10:	[B]	**Q30:**	[B]	**Q50:**	[B]	**Q70:**	[B]	**Q90:**	[A]
Q11:	[B]	**Q31:**	[B]	**Q51:**	[B]	**Q71:**	[B]	**Q91:**	[A]
Q12:	[B]	**Q32:**	[B]	**Q52:**	[B]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[B]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[B]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[B]	**Q35:**	[A]	**Q55:**	[B]	**Q75:**	[B]	**Q95:**	[A]
Q16:	[B]	**Q36:**	[C]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[B]	**Q37:**	[A]	**Q57:**	[B]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[B]	**Q38:**	[B]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[B]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[C]
Q20:	[B]	**Q40:**	[B]	**Q60:**	[C]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

Inner integral: int_0^(2a - y) x*y dx = y * [x^2 / 2]_0^(2a - y) = (y / 2) * (2a - y)^2 = (y / 2) * (4a^2 - 4ay + y^2).
Outer integral: I2 = (1/2) * int_a^(2a) (4a^2*y - 4a*y^2 + y^3) dy
= (1/2) * [2a^2*y^2 - (4a/3)*y^3 + y^4 / 4]_a^(2a).
- At y = 2a: 2a^2*(4a^2) - (4a/3)*(8a^3) + 16a^4 / 4 = 8a^4 - (32/3)a^4 + 4a^4 = 12a^4 - (32/3)a^4 = (4/3)a^4.
- At y = a: 2a^2*(a^2) - (4a/3)*(a^3) + a^4 / 4 = 2a^4 - (4/3)a^4 + a^4 / 4 = (11/12)a^4.
- Difference: (4/3)a^4 - (11/12)a^4 = (16 - 11)/12 a^4 = (5/12)a^4.
- I2 = (1/2) * (5/12)a^4 = (5/24)a^4.
5. Total Integral:
I = I1 + I2 = a^4 / 6 + (5/24)a^4 = (4/24)a^4 + (5/24)a^4 = (9/24)a^4 = (3/8)a^4.
Conclusion:
The value of the integral is (3/8) * a^4.

>>Solution Step-by-Step Question 8: Solve the improper integral using the appropriate transformation and evaluate the double integral. Use integral: ∫₀^∞ e^(-x^2) dx = √π / 2 to evaluate the improper integral: I = int

J = det([dx/dr, dx/dtheta; dy/dr, dy/dtheta]) = det([cos theta, -r*sin theta; sin theta, r*cos theta])
= r * cos^2 theta + r * sin^2 theta = r.
- By the change of variables theorem for multiple integrals:
iint_R f(x, y) dx dy = iint_{R'} f(r*cos theta, r*sin theta) * |J| dr dtheta
= iint_{R'} f(r*cos theta, r*sin theta) * r dr dtheta. (Proved).
2. Evaluation of Double Integral over First Quadrant:
- Region R: x in [0, infinity), y in [0, infinity) (first quadrant of the plane).
- In polar coordinates:
Radial coordinate r ranges from 0 to infinity.
Angular coordinate theta ranges from 0 to pi / 2.
- x^2 + y^2 = r^2.
- I = int_0^(pi/2) [int_0^infinity e^(-r^2) * r dr] dtheta.
- Inner integral: int_0^infinity e^(-r^2) * r dr:
Let u = r^2, du = 2r dr => r dr = du / 2.
int_0^infinity e^(-u) * (du / 2) = (1/2) * [-e^(-u)]_0^infinity = (1/2) * (0 - (-1)) = 1/2.
- Outer integral:
I = int_0^(pi/2) (1/2) dtheta = (1/2) * [theta]_0^(pi/2) = (1/2) * (pi / 2) = pi / 4.
3. Deduction of the Gaussian Integral:
- Notice that the double integral factors into two identical independent single integrals:
I = int_0^infinity e^(-x^2) dx * int_0^infinity e^(-y^2) dy = [int_0^infinity e^(-x^2) dx]^2.
- Since I = pi / 4:
[int_0^infinity e^(-x^2) dx]^2 = pi / 4.
- Taking the positive square root:
int_0^infinity e^(-x^2) dx = sqrt(pi) / 2.
- By symmetry of the even integrand:
int_{-infinity}^infinity e^(-x^2) dx = 2 * (sqrt(pi) / 2) = sqrt(pi).

>> Solution to Question 5: Evaluate the double integral: I = iint_R sqrt(a^2 - x^2 - y^2) dx dy, where R is the region in the first quadrant enclosed by the circle x^2 +

Geometric Meaning:
The integral represents the volume of one-eighth of a solid sphere of radius a:
V_octant = (1/8) * [(4/3) * pi * a^3] = pi * a^3 / 6. (Matches perfectly).

>>**Solution Statement:** double integration. find the area between the curve r = 2a * (1 + cos theta).

= a^2 * [(3/2)*pi + 2*sin(pi) + (1/4)*sin(2pi) - 0]
= a^2 * [(3/2)*pi + 0 + 0]
= (3/2) * pi * a^2.
Conclusion:
The area of the cardioid is (3/2) * pi * a^2.

(i) About perpendicular central axis: I_0 = (1/2) * M * R^2.
(ii) About any diameter: I_d = (1/4) * M * R^2.

>>**Solution** **Question** Evaluate the volume of the solid bounded by the paraboloid z = x^2 + y^2 and the horizontal plane

- Inner integral (with respect to z):
int_{r^2}^4 dz = 4 - r^2.
- Middle integral (with respect to r):
int_0^2 (4 - r^2) * r dr = int_0^2 (4r - r^3) dr
= [2*r^2 - r^4 / 4]_0^2 = [2(4) - 16/4] - 0 = 8 - 4 = 4.
- Outer integral (with respect to theta):
V = int_0^(2*pi) 4 dtheta = 4 * [theta]_0^(2*pi) = 4 * (2*pi) = 8*pi.
Conclusion:
The volume enclosed by the paraboloid and the plane is 8*pi.

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2. Derivation via Repeated Beta Integrals:
- Limits: z from 0 to 1 - x - y; y from 0 to 1 - x; x from 0 to 1.
- Inner integral with respect to z:
int_0^(1 - x - y) z^(n - 1) dz = (1 - x - y)^n / n.
- Middle integral with respect to y:
int_0^(1 - x) y^(m - 1) * (1 - x - y)^n dy.
Substitute y = (1 - x) * t => dy = (1 - x) dt:
int_0^1 [(1 - x)^(m-1) * t^(m-1)] * [(1 - x)^n * (1 - t)^n] * (1 - x) dt
= (1 - x)^(m + n) * int_0^1 t^(m-1) * (1 - t)^n dt
= (1 - x)^(m + n) * Beta(m, n + 1).
- Divide by n: Beta(m, n+1)/n = [Gamma(m)*Gamma(n+1)] / [n * Gamma(m+n+1)]
Since Gamma(n+1) = n*Gamma(n), this equals [Gamma(m)*Gamma(n)] / Gamma(m+n+1).
- Outermost integral with respect to x:
int_0^1 x^(l - 1) * (1 - x)^(m + n) dx = Beta(l, m + n + 1)
= [Gamma(l) * Gamma(m + n + 1)] / Gamma(l + m + n + 1).
- Multiplying both factors cancels Gamma(m + n + 1):
I = [Gamma(l) * Gamma(m) * Gamma(n)] / Gamma(l + m + n + 1). (Proved).

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1. Definition:
- The work done by a force field F along a directed curve C parametrized by r(t) for t in [t1, t2] is:
W = int_C F . dr = int_{t1}^{t2} [Fx * (dx/dt) + Fy * (dy/dt) + Fz * (dz/dt)] dt.
2. Parametric Expressions:
- x = t => dx = dt.
- y = t^2 => dy = 2t dt.
- z = t^3 => dz = 3t^2 dt.
- Limits: at (0, 0, 0), t = 0; at (1, 1, 1), t = 1.
3. Express F in terms of parameter t:
- Fx = 3x^2 + 6y = 3(t^2) + 6(t^2) = 9*t^2.
- Fy = -14yz = -14(t^2)(t^3) = -14*t^5.
- Fz = 20xz^2 = 20(t)(t^3)^2 = 20(t)(t^6) = 20*t^7.
4. Form the Dot Product F . dr:
F . dr = Fx*dx + Fy*dy + Fz*dz
= (9*t^2)*dt + (-14*t^5)*(2t dt) + (20*t^7)*(3t^2 dt)
= (9*t^2 - 28*t^6 + 60*t^9) dt.
5. Integrate from t = 0 to t = 1:
W = int_0^1 (9*t^2 - 28*t^6 + 60*t^9) dt
= [9*(t^3 / 3) - 28*(t^7 / 7) + 60*(t^10 / 10)]_0^1
= [3*t^3 - 4*t^7 + 6*t^10]_0^1
= [3(1) - 4(1) + 6(1)] - 0
= 3 - 4 + 6 = 5.
Conclusion:
The total work done by the force field is 5 units.

>>**Solution to Question 16:** a conservative vector field. Prove that F = (2xy + z^3)*i + x^2*i + 3xz^2*k is conservative, find its scalar potential function phi(x, y, z) such that F = grad(phi), and evaluate the line integral of F along any smooth curve C from (1, 0, 1) to (2, 1, 3).

- Work Done:
W = 201 - (-1) = 202.
Conclusion:
The vector field is conservative with potential phi = x^2*y + x*z^3, and the work done is 202.

- dQ/dx = d/dx [x^2] = 2x.
- dP/dy = d/dy [xy + y^2] = x + 2y.
- dQ/dx - dP/dy = 2x - (x + 2y) = x - 2y.
- In region R: for a given x in [0, 1], y ranges from the parabola y = x^2 (lower) to line y = x (upper).
- iint_R (x - 2y) dA = int_0^1 [int_{x^2}^x (x - 2y) dy] dx.
- Inner integral:

int_{x^2}^x (x - 2y) dy = [x*y - y^2]_{x^2}^x
= [x(x) - x^2] - [x(x^2) - (x^2)^2]
= [x^2 - x^2] - [x^3 - x^4] = -x^3 + x^4.

- Outer integral:

int_0^1 (-x^3 + x^4) dx = [-x^4 / 4 + x^5 / 5]_0^1 = -1/4 + 1/5 = -5/20 + 4/20 = -1/20.

Step 3: Evaluate the Line Integral oint_C (P dx + Q dy):

Boundary C consists of C1 (parabola y = x^2 from 0 to 1) and C2 (line y = x from 1 to 0):

- Along C1: y = x^2 => dy = 2x dx.

int_{C1} = int_0^1 [(x(x^2) + (x^2)^2) dx + x^2 * (2x dx)]
= int_0^1 [(x^3 + x^4) + 2x^3] dx = int_0^1 (3x^3 + x^4) dx
= [3*x^4 / 4 + x^5 / 5]_0^1 = 3/4 + 1/5 = 15/20 + 4/20 = 19/20.

- Along C2: y = x => dy = dx (x traverses from 1 to 0):

int_{C2} = int_1^0 [(x(x) + x^2) dx + x^2 dx] = int_1^0 [2x^2 + x^2] dx
= int_1^0 3x^2 dx = [x^3]_1^0 = 0 - 1 = -1 = -20/20.

- Total Line Integral:

oint_C = int_{C1} + int_{C2} = 19/20 - 20/20 = -1/20.

Conclusion:

Line Integral (-1/20) = Double Integral (-1/20). Green's Theorem is fully verified.

>>Solution to Question 19: State the line integral area formula from Green's Theorem: Area = (1/2) * oint_C (x dy - y dx). Apply this formula to find the area enclos

= 3*pi * a^2 / 8.
Conclusion:
The area enclosed by the astroid is (3/8) * pi * a^2.

- On S1: (curl F) . n = k . k = 1.
- iint_{S1} (curl F) . n dS = iint_{S1} 1 dA = Area of unit disk = pi * (1)^2 = pi.

Step 3: Evaluate the Line Integral oint_C F . dr:

- Boundary C is the circle x^2 + y^2 = 1, z = 0.
- Parametric equations: x = cos t, y = sin t, z = 0 (t in [0, 2*pi]).
- Differentials: dx = -sin t dt, dy = cos t dt, dz = 0.
- F on C (with z = 0):

Fx = 2x - y = 2*cos t - sin t.

Fy = -y*z^2 = 0.

Fz = -y^2*z = 0.

- F . dr = Fx * dx = (2*cos t - sin t) * (-sin t dt)

= (-2*sin t * cos t + sin^2 t) dt

= [-sin(2t) + (1 - cos 2t)/2] dt.

- Integrate from 0 to 2*pi:

oint_C F . dr = int_0^(2*pi) [-sin(2t) + 1/2 - (1/2)*cos(2t)] dt

= [(1/2)*cos(2t) + t/2 - (1/4)*sin(2t)]_0^(2*pi)

= [(1/2) + pi - 0] - [1/2 + 0 - 0] = pi.

Conclusion:

Surface Integral (pi) = Line Integral (pi). Stokes' Theorem is completely verified.

>>Solution to Question 22: Gauss' Divergence Theorem. Provide a rigorous analytical proof for a simple 3D volume V bounded by an orientable closed surfa

three equations yields:
iint_S [F1*(n.i) + F2*(n.j) + F3*(n.k)] dS = iiint_V [dF1/dx + dF2/dy + dF3/dz] dV
=> iint_S F . n dS = iiint_V div(F) dV.
Hence proved.

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div(F) = d/dx(x^3) + d/dy(y^3) + d/dz(z^3) = 3x^2 + 3y^2 + 3z^2
= 3 * (x^2 + y^2 + z^2).
Step 2: Apply Gauss' Divergence Theorem:
iint_S F . n dS = iiint_V div(F) dV
= 3 * iiint_V (x^2 + y^2 + z^2) dV,
where V is the solid ball of radius a.
Step 3: Transform to Spherical Polar Coordinates:
- x^2 + y^2 + z^2 = rho^2.
- dV = rho^2 * sin phi drho dphi dtheta.
- Limits: rho in [0, a], phi in [0, pi], theta in [0, 2*pi].
Step 4: Evaluate the Triple Integral:
iiint_V (x^2 + y^2 + z^2) dV = int_0^(2*pi) dtheta * int_0^pi sin phi dphi * int_0^a rho^2 *
(rho^2 drho)
= [2*pi] * [2] * [int_0^a rho^4 drho]
= 4*pi * [rho^5 / 5]_0^a
= 4*pi * a^5 / 5 = (4/5) * pi * a^5.
Step 5: Multiply by 3:
iint_S F . n dS = 3 * [(4/5) * pi * a^5]
= (12/5) * pi * a^5.
Conclusion:
The total outward flux through the sphere is (12/5) * pi * a^5.

>> Solution to Question 25: Evaluate the surface integral iint_S F . n dS, where F = z*i + x*i - 3y^2*z*k and S is the surface of the cylinder x^2 + y^2 = 16 included i

Conclusion:
The value of the surface integral is 90.

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iiint_V (drho / dt) dV = - iint_S (rho * v) . n dS.
3. Application of Gauss' Divergence Theorem:
- Transform the surface integral into a volume integral:
iint_S (rho * v) . n dS = iiint_V div(rho * v) dV.
- Substitute:
iiint_V (drho / dt) dV = - iiint_V div(rho * v) dV
=> iiint_V [drho / dt + div(rho * v)] dV = 0.
- Since volume V is completely arbitrary, the integrand must vanish identically everywhere:
drho / dt + div(rho * v) = 0 (The Continuity Equation).
4. Incompressible Fluid Condition:
- For an incompressible liquid broth, fluid density rho is strictly constant (drho/dt = 0 and grad(rho) = 0):
div(rho * v) = rho * div(v) = 0.
- Since rho != 0:
div(v) = 0. (Hence proved).
Conclusion:
The velocity field of an incompressible fluid is strictly solenoidal.

>> Solution to Question 28: In biotechnology bioreactor design, use surface flux integration to calculate the total oxygen mass transfer rate across spherical gas bu

iint_S J . n dS = - iiint_V D * del^2 C dV.

4. Equating to Metabolic Consumption:

- - iint_S J . n dS = iiint_V D * del^2 C dV = iiint_V R_rxn dV

=> iiint_V [D * del^2 C - R_rxn] dV = 0.

- Since this balance holds for any arbitrary sub-volume of the cell:

D * del^2 C = R_rxn,

which is the fundamental steady-state reaction-diffusion equation (Poisson's equation for biotechnology).

Conclusion:

The integral of the outward diffusive flux over the cell membrane exactly balances the total volumetric enzymatic reaction within the cell: iint_S (-D * grad C) . n dS = - iiint_V R_rxn dV.
